

Nested loops

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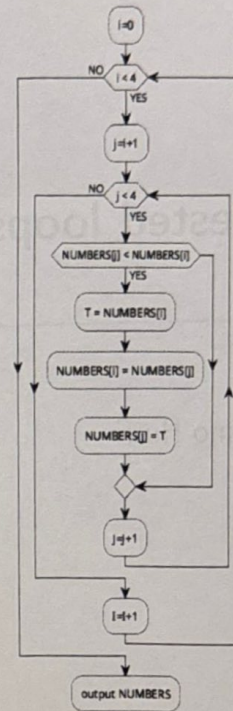
Example 1

Write a Python program with positive integer constant N , printing N rows with integers $1, 2, \dots, k$ on each row, with $k=N$ on first row, and with k decreasing by one on each consecutive row. The output below is for $N=7$.

```
1 2 3 4 5 6 7
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Example 2

1. Trace the program when `numbers=[7, -5, 11, 6]`. (See next page.)
2. What is the purpose of the program?



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Example continues

i	i<4	j	j<4	NUMBERS[j] < NUMBERS[i]	NUMBERS

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Example 3

Write a Python program which

- contains a list of flights (strings) and a list of airline codes (strings)
- prints flights and airline codes
- prints flights matching airline codes.

Sample output:

```
flights: ['AY664', 'BA047', 'AF110', 'LH554', 'AY101']
airlines: ['AY', 'UA', 'LH']
AY: AY664 AY101
UA:
LH: LH554
```

Homework 1

Write a program that produces the following output.

1.

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

2.

```
1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9
```

Homework 2

In the program, A and B are lists of integers with 5 and 6 elements, respectively.

1. What is the purpose of the program?
2. Why is this program not very efficient?
3. Implement an efficient version in Python for lists of arbitrary lengths.

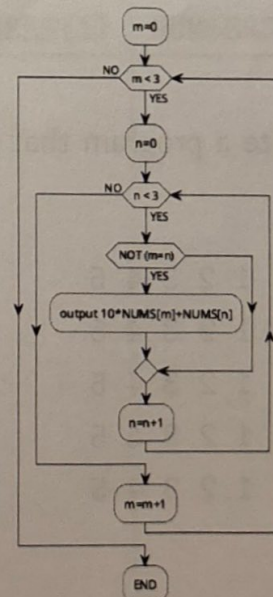
```
flag=False
loop n from 0 to 4
  loop m from 0 to 5
    if A[n]=B[m]
      flag=True
    end if
  end loop
end loop

output flag
```

Homework 3

The input to the program is integer list NUMS with 3 elements.

1. Trace the program using the table on next page when NUMS=[3, 7, 4].
2. What is the purpose of the program.
3. Write a corresponding Python program that works on array of any length.



Homework continues

m	m<3	n	n<3	NOT(m=n)	output

Homework 4

Write a Python program that produces the following output.

1.

* _ _ _ _ _

**- - - -

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*****_

2.

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--*

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Homework 5

Write a function

```
uniques(numbers)
```

which returns True if all the elements of integer list `numbers` are unique and False otherwise. Test your method in a program.

Homework 6

Write a function

```
symm_diff(a, b)
```

which returns the *symmetric difference* of integer lists `a` and `b`: a list with those elements of `a` that are not elements of `b`, followed by those elements of `b` that are not elements of `a`. Example use in a program:

```
a: [4, 4, 6, 11, -2, 3]
```

```
b: [5, 11, 11, -3, 3, 5]
```

```
symm_diff: [4, 4, 6, -2, 5, -3, 5]
```